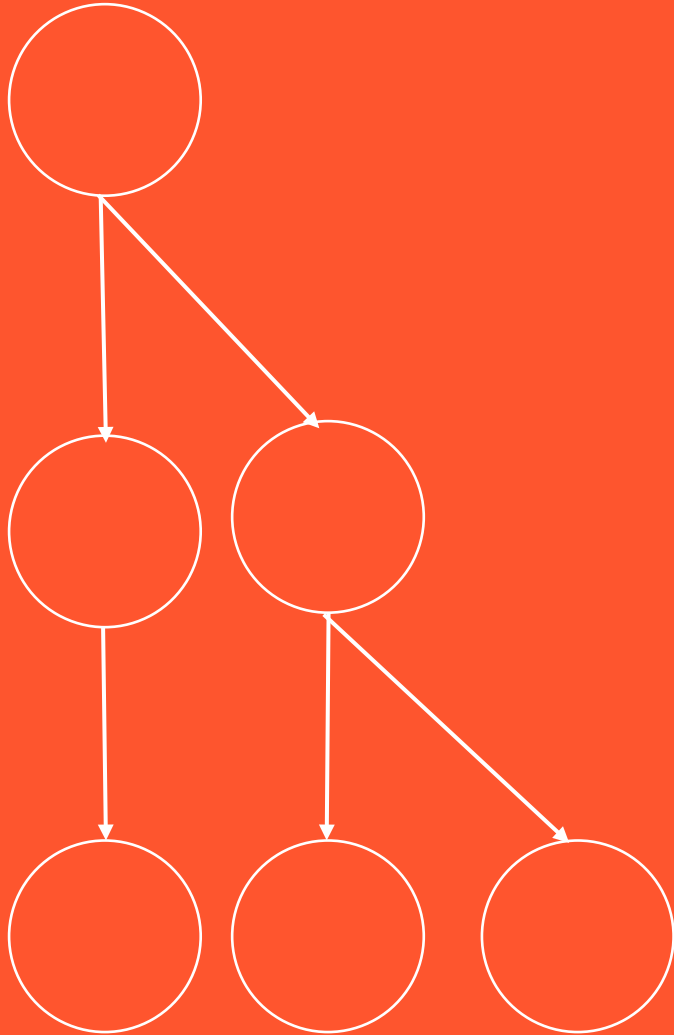


BCOG 100

Brains and Reinforcement Learning

Lecture 3: The Implementational Level



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URBANA-CHAMPAIGN

Reading Check 6-3

Password: critic

Question: operant conditioning isn't quite the same thing as reinforcement learning, as we discussed last time. What's the difference?



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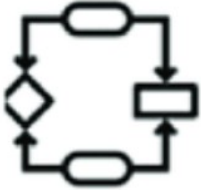


Computation

What does the system do?

Recap & Overview

Error-driven learning, TD, actor-critic



Algorithm

How does the system do it?

Today:

- How the brain implements RL
- Focus: dopamine and basal ganglia circuit



Implementation

How is the system realized?

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		Arousal	
		High	Low
Valence	Good	Excited	Content
	Bad	Anxious	Depressed

Evolutionary Roots of Neuromodulators

- Classical: glutamate & GABA (synaptic signals)
- Neuromodulators: dopamine, serotonin, norepinephrine (state control)
- Together shaped primitive valence & arousal

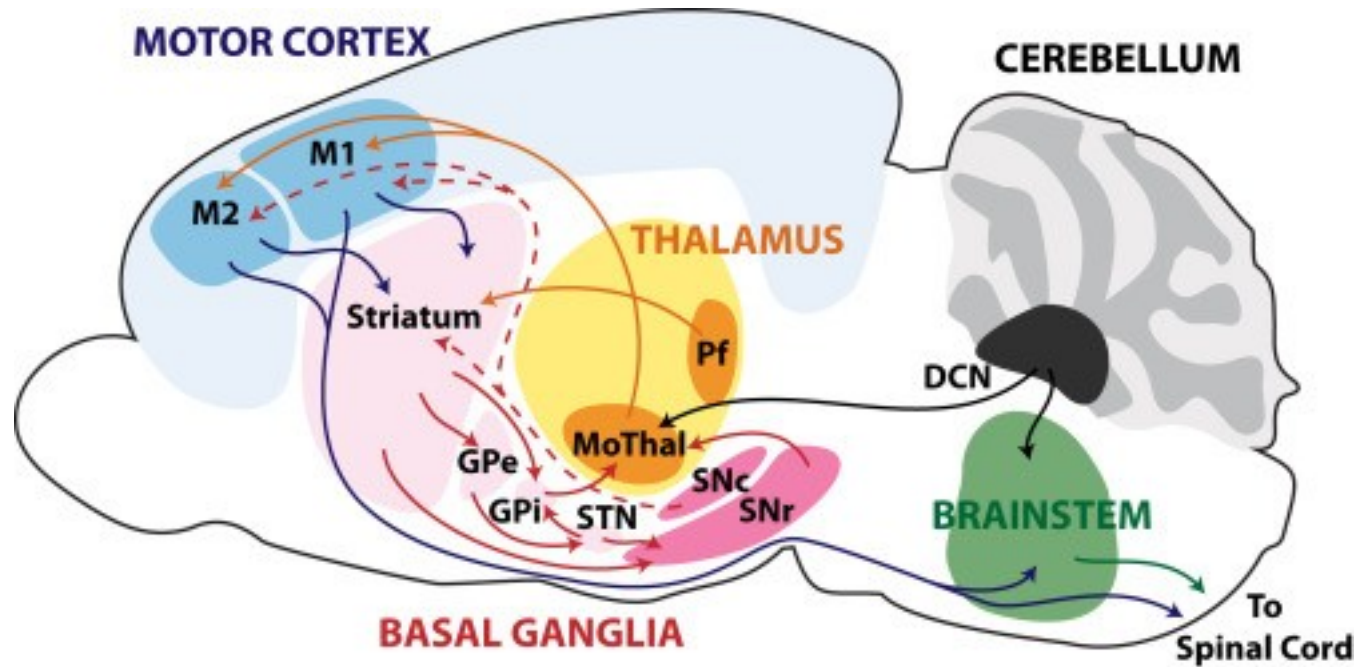
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		Arousal	
		High	Low
Valence	Good	Excited	Content
	Bad	Anxious	Depressed

Dopamine Before RL

- Ancient neuromodulator in invertebrates
- Tunes arousal, movement, motivation
- Effect depends on context and circuit

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The Evolutionary Repurposing of Dopamine

- Basal ganglia: global action gate
- Default: inhibit all actions until released
- Dopamine repurposed into teaching signals

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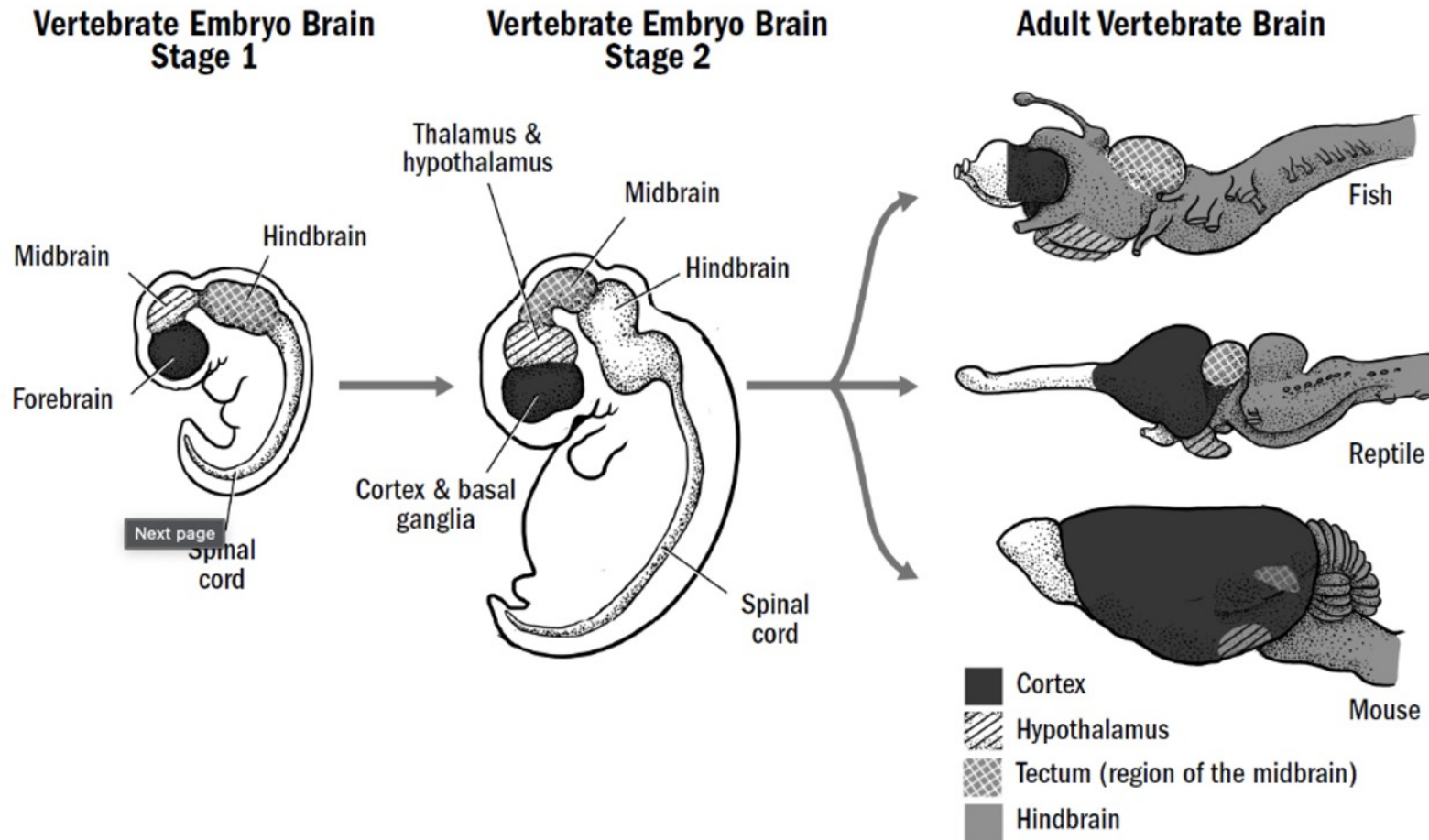
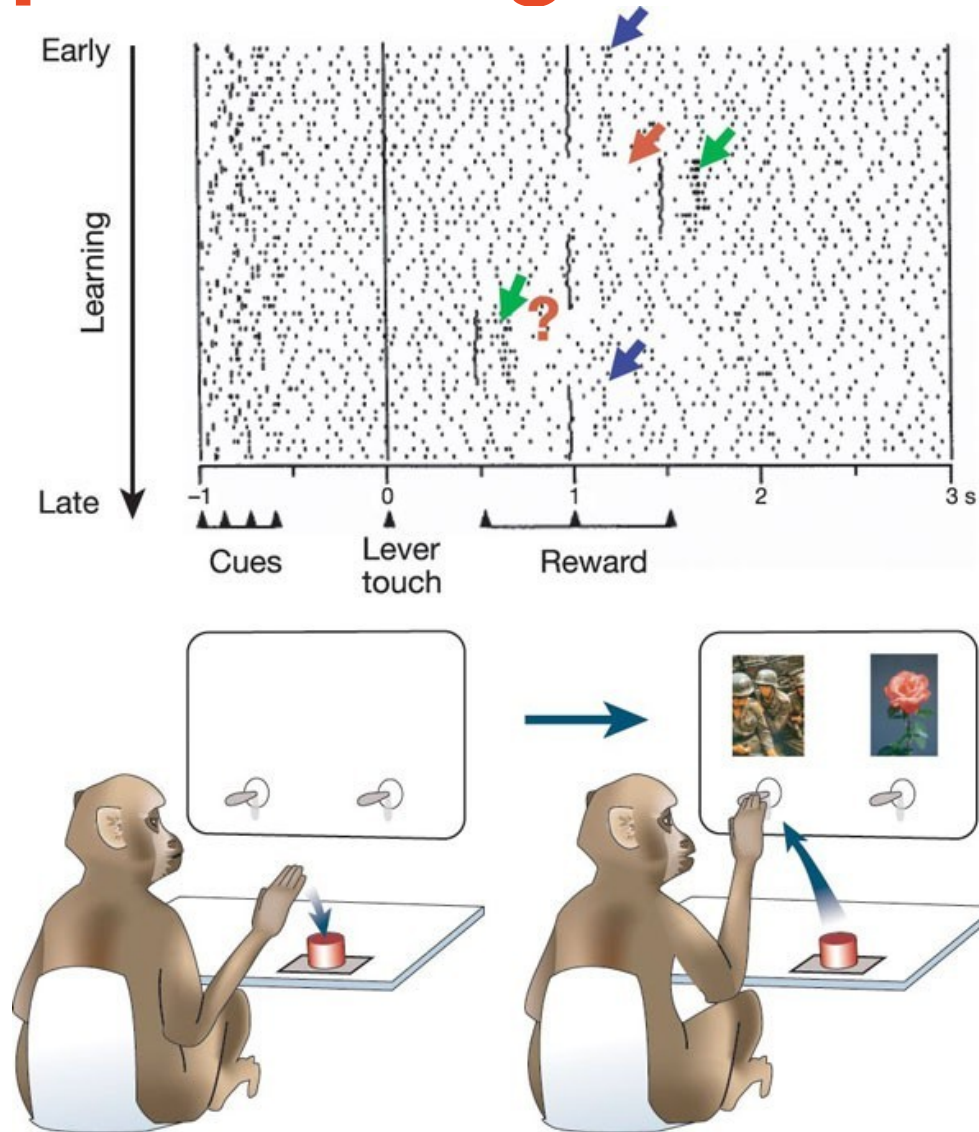


Figure 5.3: The shared embryonic development of vertebrates

Brains Built for Reinforcement Learning

- Basal ganglia conserved across vertebrates
- Cortex → striatum → pallidum → thalamus → cortex
- Action gating + dopamine teaching = RL engine

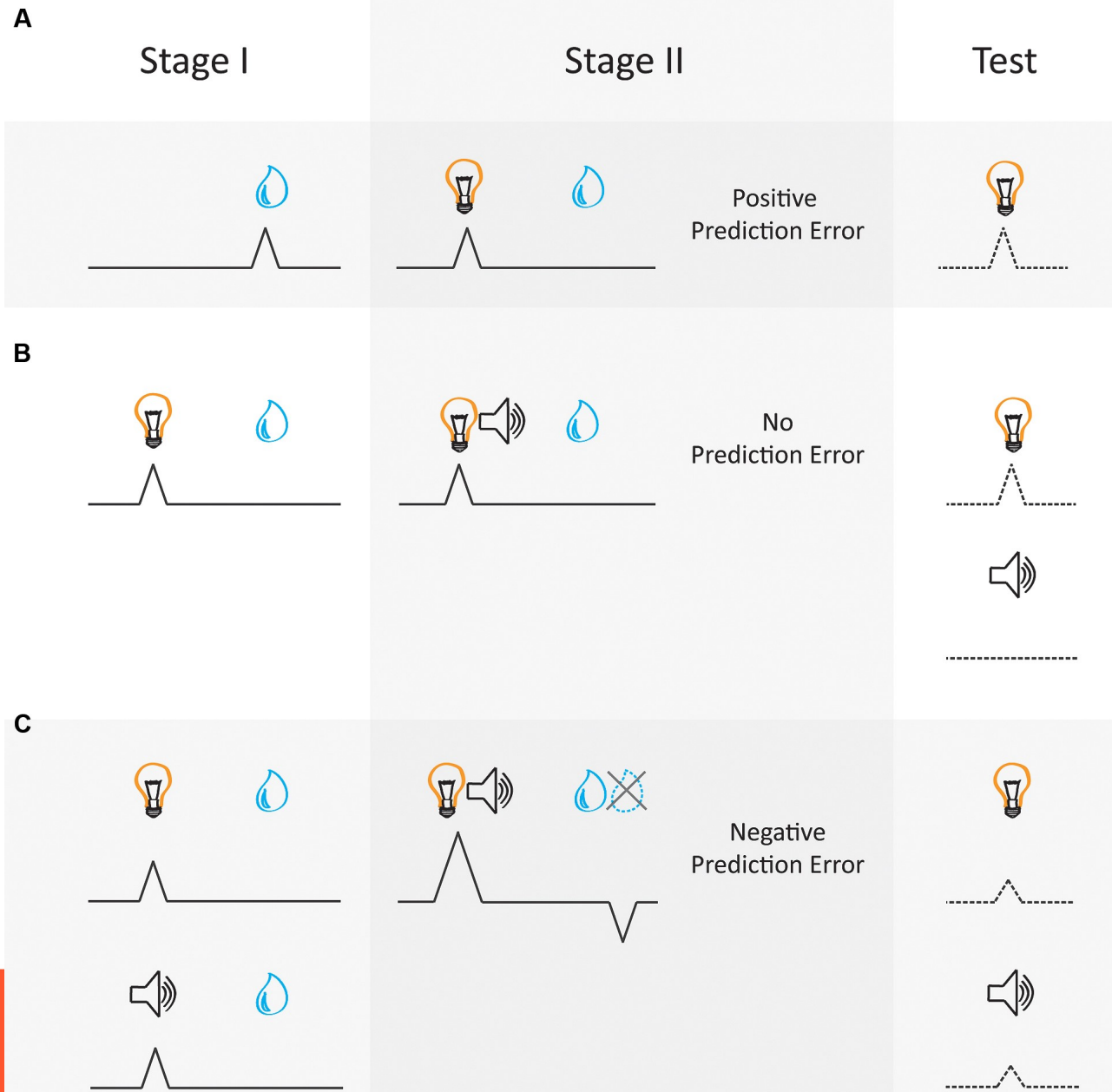
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Dopamine: First Clues

- Schultz's monkey experiments (1990s)
- Unexpected reward → dopamine burst
- With training, response shifts to cue
- Reward omitted → dopamine dip

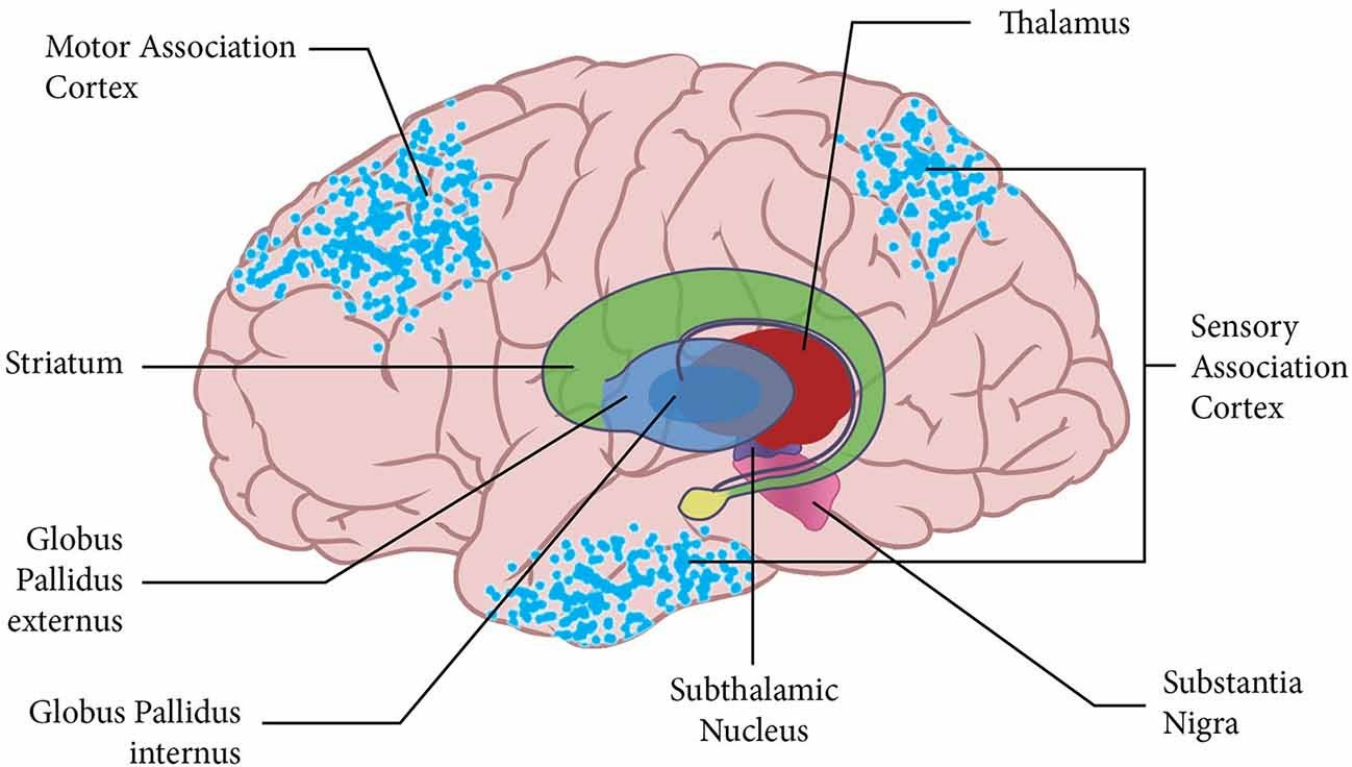
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Dopamine as Prediction Error

- Phasic firing matches TD learning rules
- Shifts from reward to predictive cue
- Blocking and over-expectation also follow error laws

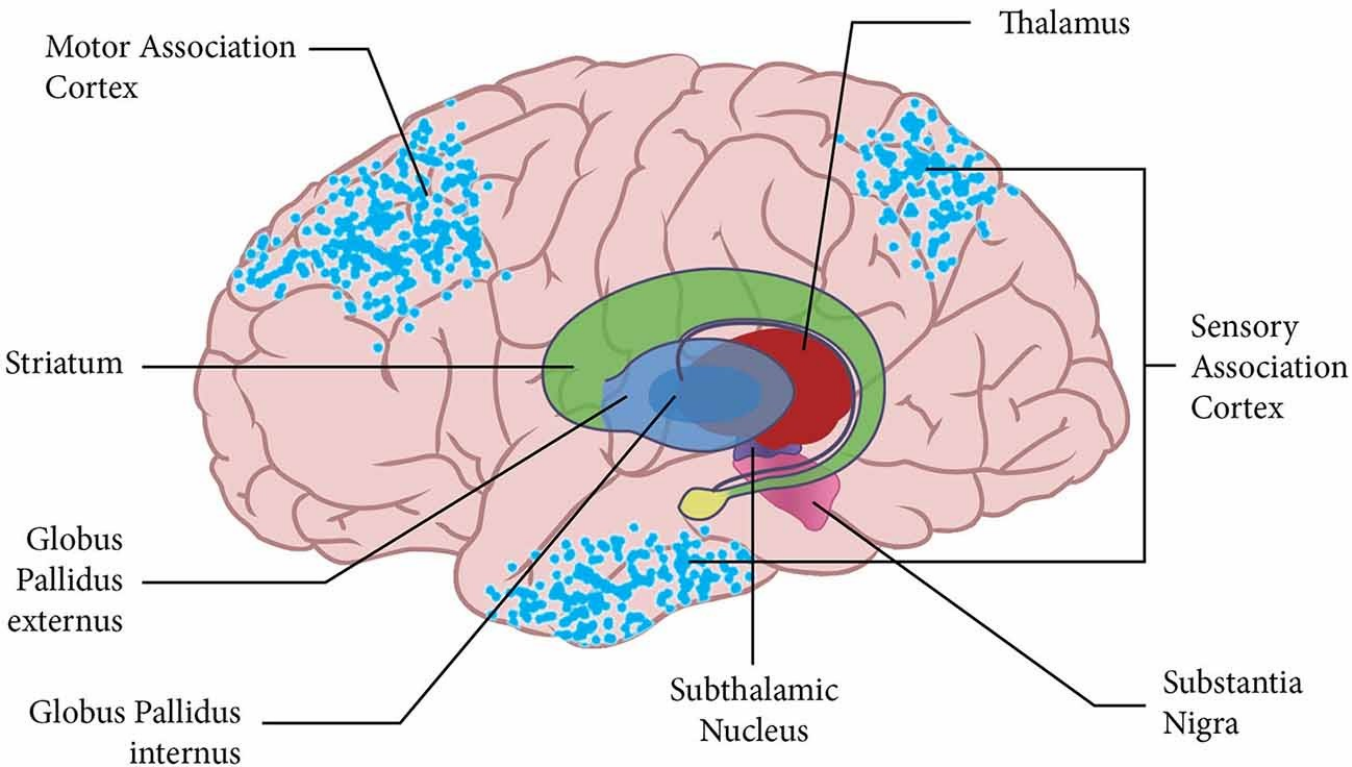
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Basal Ganglia Anatomy

- Input hub: striatum
- Output nuclei: globus pallidus, substantia nigra
- Loop: cortex → striatum → pallidum → thalamus → cortex

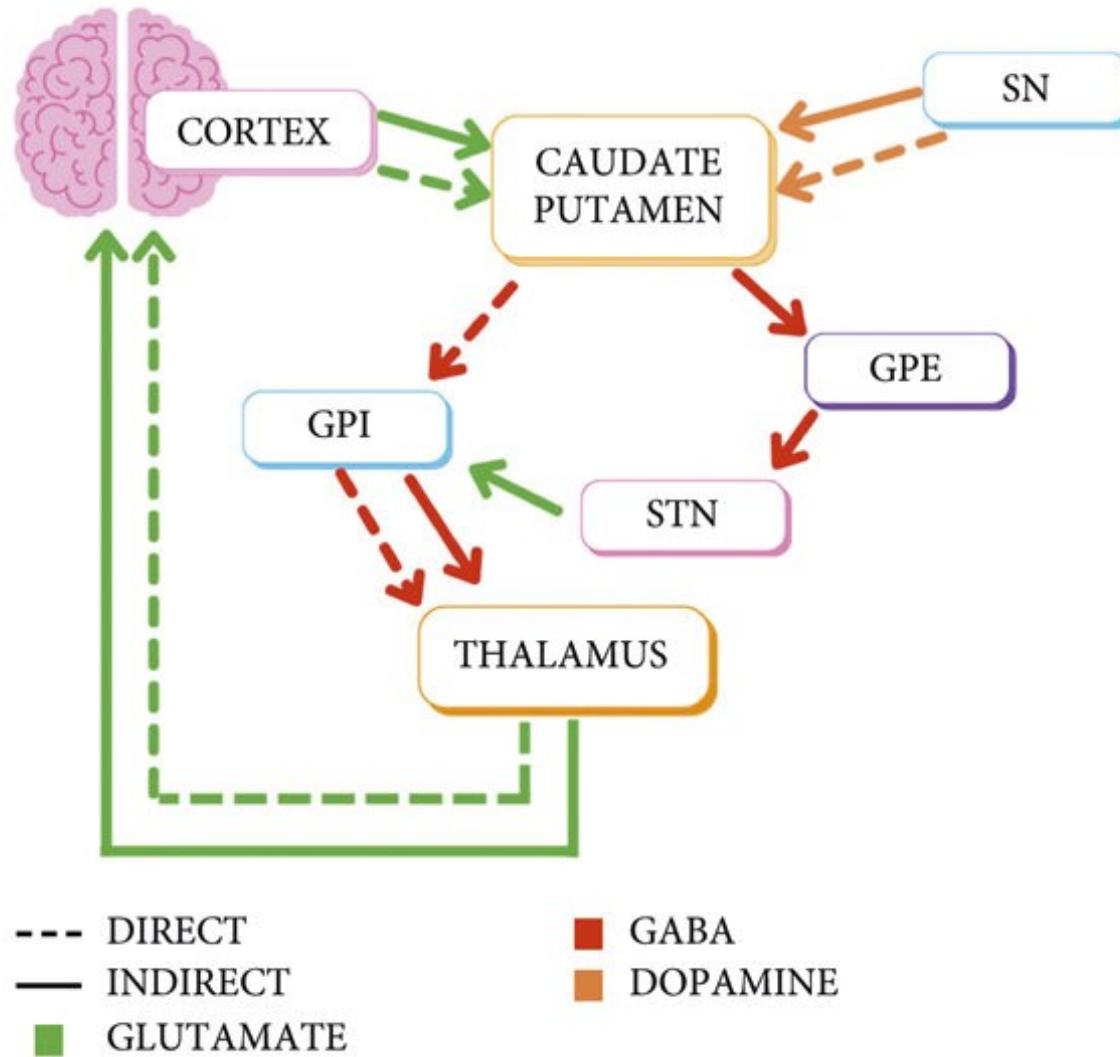
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Actor-Critic in the Brain

- Ventral striatum $\approx$ critic (value predictions)
- Dorsal striatum $\approx$ actor (action choices)
- Dopamine TD errors update both

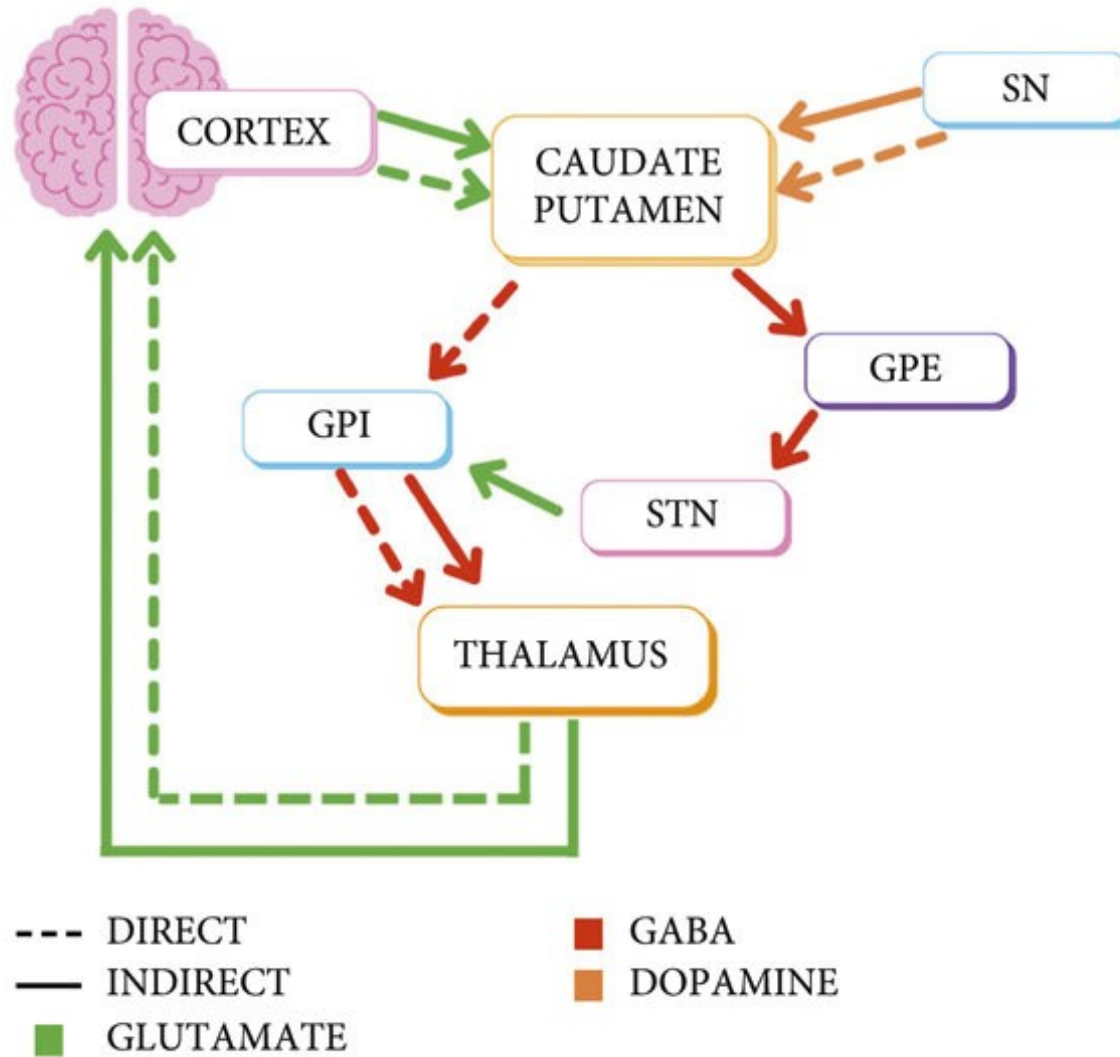
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Direct and Indirect Pathways

- D1 receptors: direct “Go” pathway → promote action
- D2 receptors: indirect “No-Go” pathway → suppress action
- Dopamine shifts balance between Go and No-Go

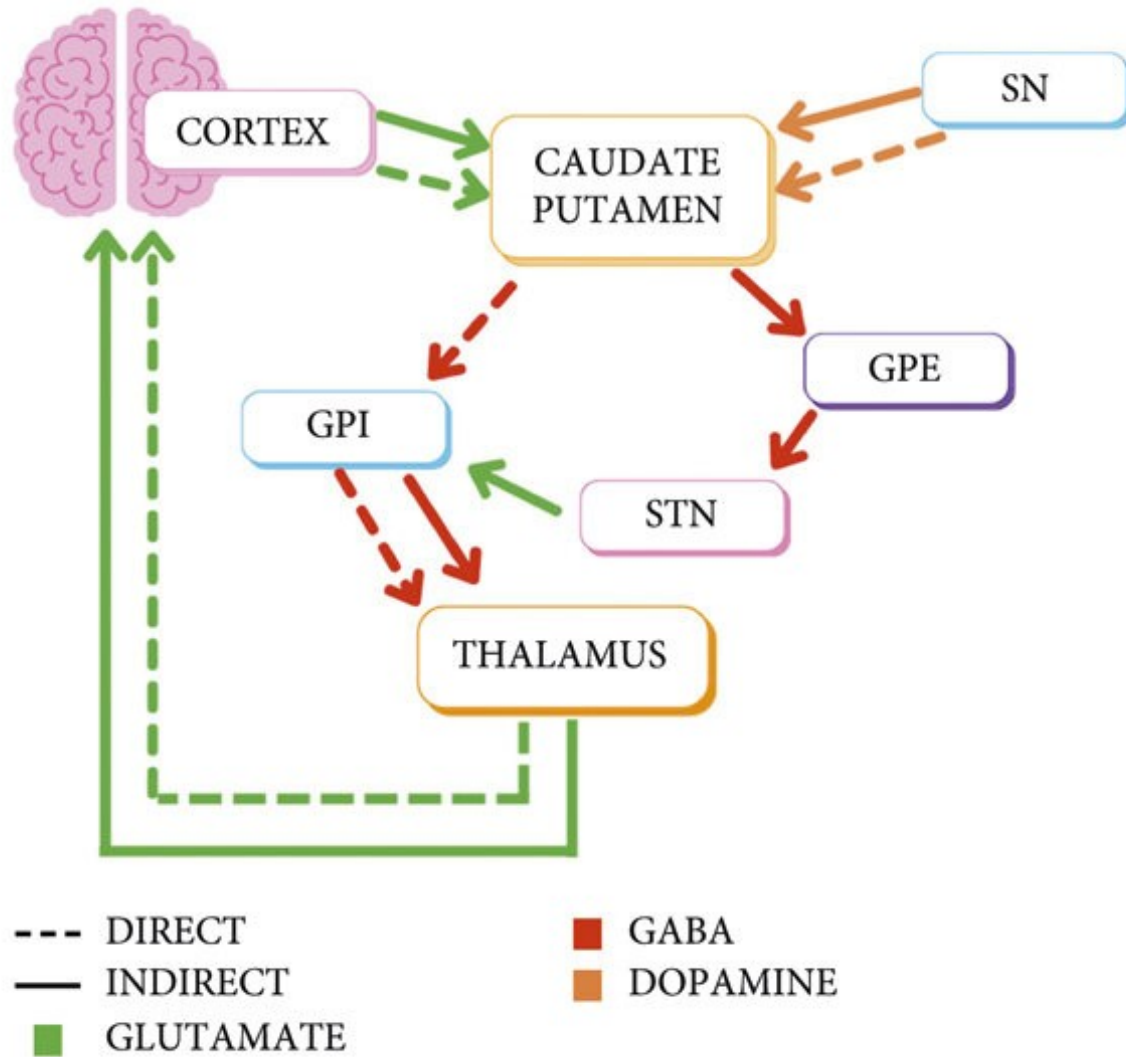
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Learning Mechanism

- Dopamine bursts → strengthen “Go” synapses (D1)
- Dopamine dips → strengthen “No-Go” synapses (D2)
- Implements trial-and-error learning in the striatum

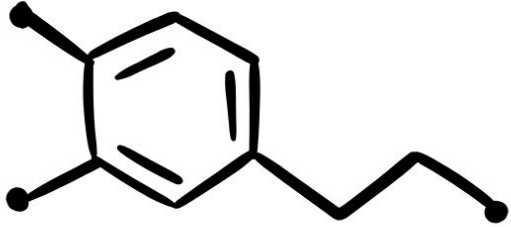
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Example: Checkers in the Brain

- Critic: ventral striatum predicts board states
- Actor: dorsal striatum selects moves
- Dopamine bursts/dips update both via plasticity

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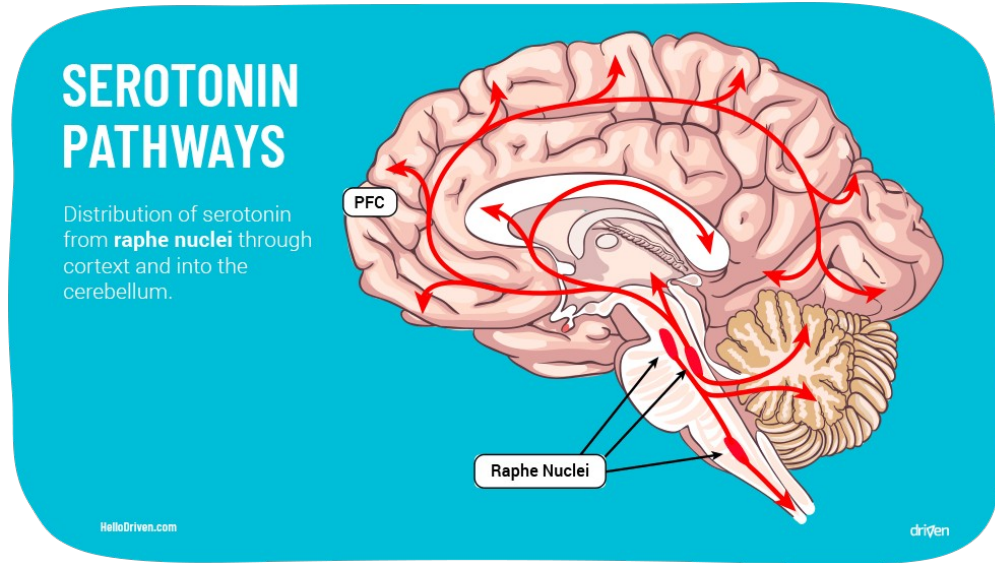
**DOPAMINE
DEBUNKED**



Beyond Simple Rewards

- Dopamine ≠ “pleasure chemical”
- Also signals novelty, salience, uncertainty
- General teaching signal, not just reward

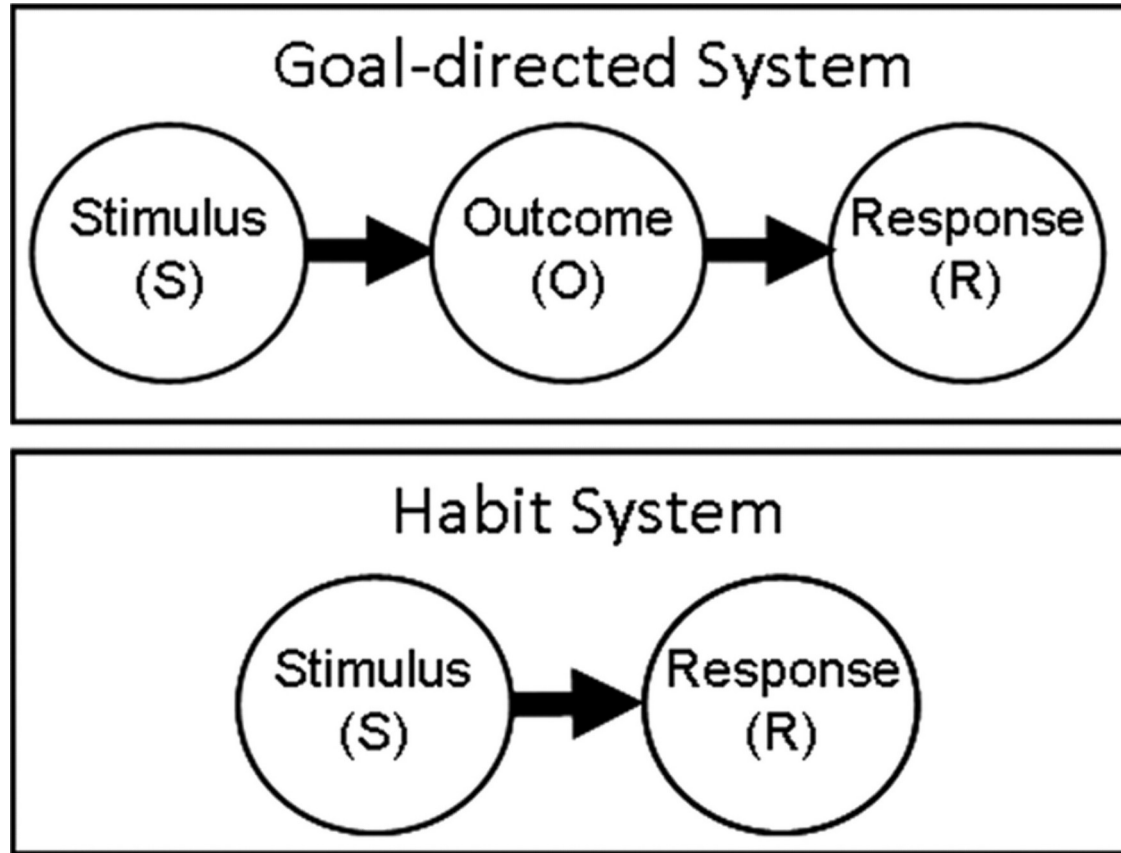
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Other Neuromodulators

- Serotonin → patience, inhibition, aversive learning
- Norepinephrine → arousal, attention, surprise
- Work with dopamine to balance behavior

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Habits vs Goal-Directed Systems

- Dorsomedial striatum → goal-directed (devaluation-sensitive)
- Dorsolateral striatum → habits (outcome-insensitive)
- Practice shifts control from goals to habits

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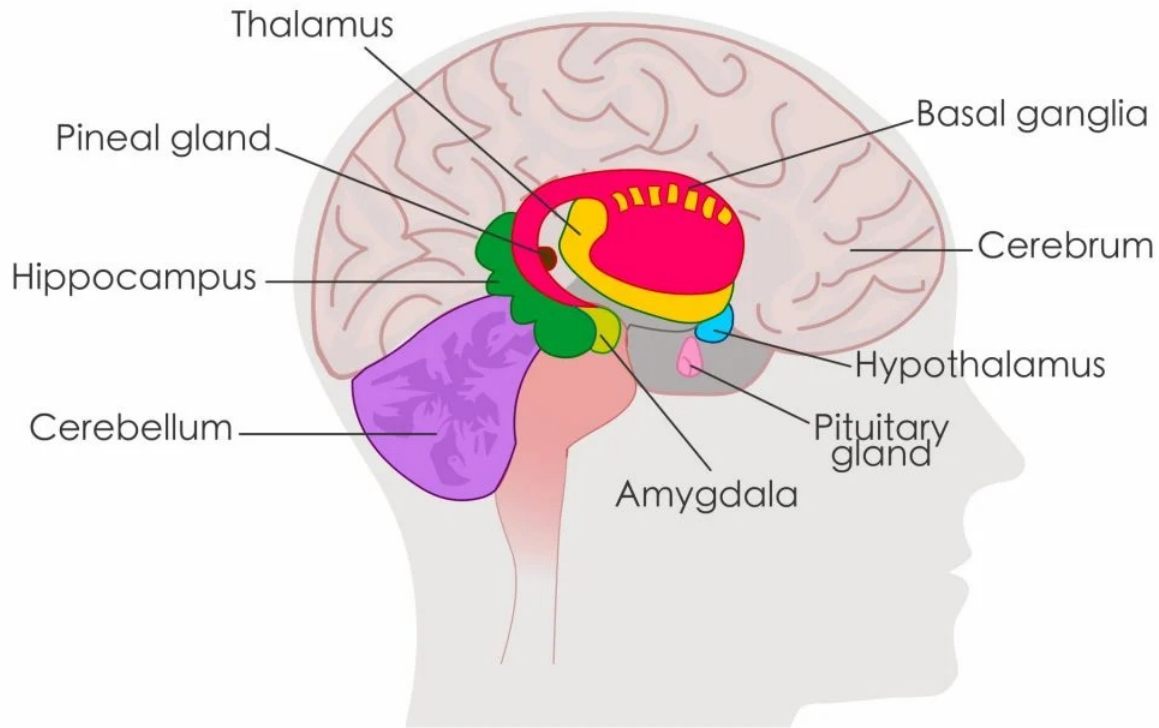


Addiction

- Drugs hijack dopamine prediction error
- Produce artificial bursts, regardless of outcome
- Shift behavior from goal-directed to habitual

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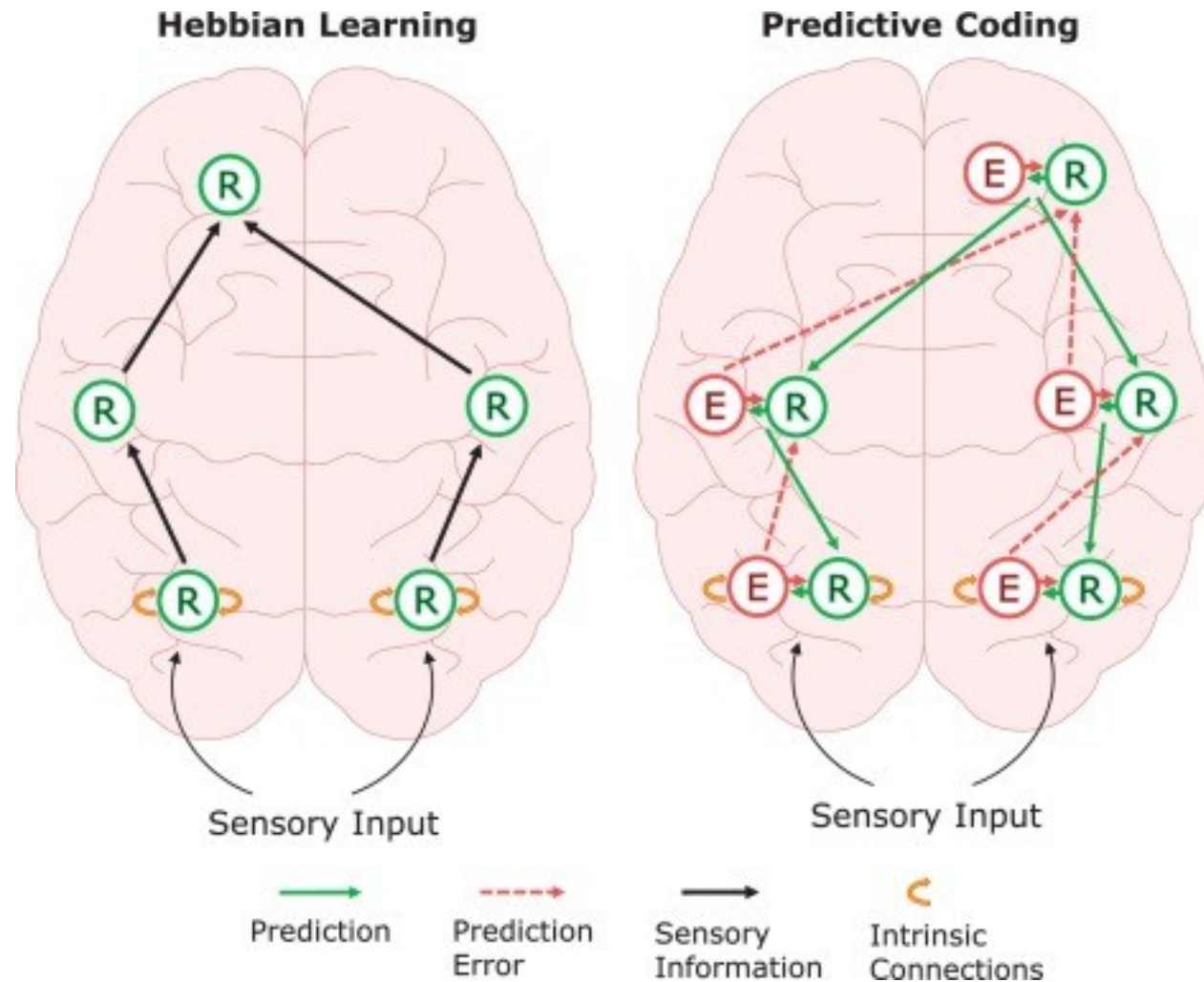
Limbic System



Other Circuits in RL

- Prefrontal cortex → planning, model-based RL
- Amygdala → valence, Pavlovian responses
- Hippocampus → context, episodic memory

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Putting It Together

- Cortex $\leftrightarrow$ basal ganglia $\leftrightarrow$ thalamus loop
- Dopamine: prediction error signal
- Actor-critic roles mapped onto brain circuits

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Evidence & Speculation

- Dopamine neurons = clear prediction error signals (Schultz, 1990s → today)
- Basal ganglia loops conserved across vertebrates
- Evolutionary “repurposing” story is plausible but inferred

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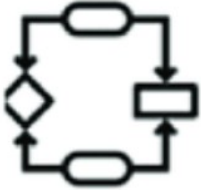


Computation

What does the system do?

What We Learned

- Dopamine: prediction error → teaching signal
- Basal ganglia: action gating + RL engine



Algorithm

How does the system do it?

What's Next

How brains and machines recognize complex patterns



Implementation

How is the system realized?